

Test - Functions

Time Allowed : 1 hour _____ Maximum Marks : 100

Instructions

1. The test consists of two Parts.

Part I

2. All questions in Part I carry +5 marks for a correct
3. Question number 1 to 7 are single answer MCQ's . Each correct response carries +5 marks while for each incorrect response, 2 marks will be deducted.
4. Question number 8 and 9 are MCQ's with none, one or more than one correct answers. Each correct response carries +5 marks while for each incorrect response, 1 mark will be deducted.
5. Question number 10 is Assertion Reason type Question. A correct response carries +5 marks while 1 mark will be deducted for an incorrect response.

Part II

6. Question number 11 to 16 are single answer MCQ's based on paragraphs. Each correct response carries +5 marks while for each incorrect response, 1 mark will be deducted.
7. Question number 17 is a Matrix match type problem. For all correct rows +20 marks will be awarded , for one row incorrect, +10 marks will be awarded and for more than rows incorrect, no credit will be given.

Part I

Q1: The complete solution set of $\frac{1}{\left|\frac{4}{x} - 10\right|} < 3$ is

- a) $x \in \mathbb{R} - \left(\frac{12}{31}, \frac{12}{29}\right)$
- b) $x \in \left(\frac{12}{31}, \frac{12}{29}\right) - \left\{\frac{2}{5}\right\}$
- c) $x \in \mathbb{R} - \{0\} - \left[\frac{12}{31}, \frac{12}{29}\right]$
- d) None of these

Q2: The point in the co-ordinate plane satisfying $|7x + 5| = -|3 - 3y|$ is

- a) $(-2, 3)$
- b) $\left(-\frac{5}{7}, 1\right)$
- c) $(-5, -3)$
- d) None of these

Q3: The complete solution set of $|\cot x| = \cot x + 2\sqrt{3}$ is

- a) $\{x : x = n\pi + \frac{\pi}{3}, n \in \mathbb{Z}\}$
- b) $\{x : x = n\pi - \frac{\pi}{3}, n \in \mathbb{Z}\}$
- c) $\{x : x = n\pi + \frac{\pi}{6}, n \in \mathbb{Z}\}$
- d) $\{x : x = n\pi - \frac{\pi}{6}, n \in \mathbb{Z}\}$

Q4: The area bounded by the curves given by $|x| + |2y| = 2$ is

- a) 2 units
- b) 4 units
- c) 6 units
- d) 8 units

Q5: A function $f : R \rightarrow [1, 2]$ given by $f(x) = \log_3(9 - 6|\cos x|)$ is

- a) One-one only
- b) Onto only
- c) One-one and onto
- d) None of these

Q6: Let $f : [0, \infty) \rightarrow S$ defined as $f(x) = 2^{x(1-x)}$ be onto, then the set S is

- a) $[0, \infty)$
- b) $[1, \infty)$
- c) $\left[2^{-\frac{1}{4}}, 1\right]$
- d) $\left[2^{-\frac{1}{4}}, \infty\right)$

Q7: A function on $f : [2, 4) \rightarrow (-\infty, 0]$ is given by $f(x) = -2 + \log_2(-x^2 + 4x)$. Then $f^{-1}(x)$ is given by

- a) $2 \pm 2\sqrt{1 - 2^x}$
- b) $2 - 2\sqrt{1 - 2^x}$
- c) $2 + 2\sqrt{4 - 2^x}$
- d) None of these

Q8: The solution set of $2|\tan x| = \tan x + \sqrt{3}$ contains

- a) $\{x : x = n\pi + \frac{\pi}{3}, n \in \mathbb{Z}\}$
- b) $\{x : x = n\pi - \frac{\pi}{3}, n \in \mathbb{Z}\}$
- c) $\{x : x = n\pi + \frac{\pi}{6}, n \in \mathbb{Z}\}$
- d) $\{x : x = n\pi - \frac{\pi}{6}, n \in \mathbb{Z}\}$

Q9. Which of the following are rational numbers.

- a) $\frac{\sqrt{3}}{2 + \sqrt{3}}$
- b) $\frac{2}{6}$
- c) $(\sqrt{5} - 2)(\sqrt{5} + 2)$
- d) $\frac{11\pi}{7\pi}$

Q10 Assertion-Reason Type Questions

Each question in this section has four choices (a),(b),(c) and (d) out of which only one is correct. Mark your choices as follows:

- (a) STATEMENT-1 is True, STATEMENT-2 is True, STATEMENT-2 is the correct explanation of STATEMENT-1.
- (b) STATEMENT-1 is True, STATEMENT-2 is True, STATEMENT-2 is **NOT** the correct explanation of STATEMENT-1.
- (c) STATEMENT-1 is True, STATEMENT-2 is False.
- (d) STATEMENT-1 is False, STATEMENT-2 is True

Q1: STATEMENT-1: For every rational number r satisfying $r^2 < 2$ one can always find a larger rational number $r + k$ ($k > 0$) for which $(r + k)^2 < 2$.

STATEMENT-2: There exist infinite rational numbers between any two rational numbers.

Part II

Comprehension 1: The first 3 terms of an A.P. are $|3x|$, $|3x - 1|$ and $|3x + 1|$. Based on this information, give the answers to the questions below

Q11: The value of x is

- a) 1
- b) 2
- c) -1
- d) None of these

Q12: The 10th term of the A.P. is

- a) $\frac{1}{4}$
- b) $\frac{5}{4}$
- c) $\frac{19}{4}$
- d) None of these

Q13: The sum of first 15 terms of the series is

- a) $\frac{225}{4}$
- b) $\frac{359}{4}$
- c) $\frac{433}{4}$
- d) None of these

Comprehension 2: A function $f_n(x)$ is defined for all $n \in \mathbb{N}$ and $f_{n+m}(x)$ is defined as $f_{n+m}(x) = f_n(f_m(x))$ where $f_1(x) = \frac{2x-1}{x+1}$ for $x \in \mathbb{R} - \{-1\}$. Using this definition, $f_2(x) = f_1(f_1(x)) = \frac{x-1}{x}$ for $x \in \mathbb{R} - \{-1, 0\}$. Similarly $f_3(x) = f_1(f_2(x)) = \frac{2-x}{1-2x}$ for $x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}\right\}$ and so on. Based on the information, answer the questions below.

Q14: The domain of definition of $g(x) = |\ln(f_{34}(x))|$ is

- a) $(-\infty, 1) - \left\{-1, 0, \frac{1}{2}\right\}$
- b) $(-\infty, 1] - \left\{-1, 0, \frac{1}{2}\right\}$
- c) $(1, \infty) - \{2\}$
- d) None of these

Q15: The complete solution set of $|f_{71}(x)| > \left|\frac{1}{f_{73}(x)}\right|$ is

- a) $(-1, 1)$
- b) $\mathbb{R} - \{2\} - [-1, 1]$
- c) $\mathbb{R} - \{-1, 1, 2\}$
- d) None of these

Q16: The values of x for which $|f_{56}(x)| > f_{56}(x)$ belong to the interval

- a) $\left(\frac{1}{2}, 2\right)$
- b) $(2, \infty)$
- c) $(0, 1)$
- d) None of these

Q17: Under Column I , some equations are given . Under Column II, some solutions satisfying some of the equations are given. Match the entry in Column I with the solution satisfying it in Column II. {Note: $[]$ is the Greatest Integer Function}

Column I

(P) $|\sin^{-1} x| = \sin^{-1} x + \frac{\pi}{6}$

(Q) $\ln |x| = |\ln x|$

(R) $|[x]| = [x]$

(S) $|x^2 - 3x + 2| > x^2 - 3x + 2$

Column II

(A) $-\frac{1}{2}$

(B) 1

(C) $\frac{3}{2}$

(D) 2

Functions Test

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D

8	A	B	C	D
9	A	B	C	D

10	A	B	C	D	E
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11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D

17				
P	A	B	C	D
Q	A	B	C	D
R	A	B	C	D
S	A	B	C	D

Functions Test

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D

8	A	B	C	D
9	A	B	C	D

10	A	B	C	D	E
----	---	---	---	---	---

11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D

17				
P	A	B	C	D
Q	A	B	C	D
R	A	B	C	D
S	A	B	C	D

Test - Functions

Q1) C

$$\{\text{Hint: } \frac{1}{\left|\frac{4}{x} - 10\right|} < 3$$

$$\Rightarrow \left|\frac{4}{x} - 10\right| > \frac{1}{3}$$

$$\text{Case I : } \frac{4}{x} - 10 > \frac{1}{3}$$

$$\Rightarrow \frac{4}{x} > \frac{31}{3}$$

$$\Rightarrow \frac{1}{x} > \frac{31}{12}$$

$$\Rightarrow 0 < x < \frac{12}{31}$$

$$\text{Case II: } \frac{4}{x} - 10 < -\frac{1}{3}$$

$$\Rightarrow \frac{4}{x} < \frac{29}{3}$$

$$\Rightarrow \frac{1}{x} < \frac{29}{12}$$

$$\Rightarrow \frac{1}{x} < 0 \text{ or } 0 < \frac{1}{x} < \frac{29}{12}$$

$$\Rightarrow x < 0 \text{ or } x > \frac{12}{29}$$

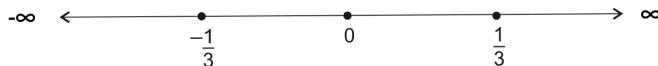
Combining all cases, we get $x \in (-\infty, 0) \cup \left(0, \frac{12}{31}\right) \cup \left(\frac{12}{29}, \infty\right)$ i.e. $x \in \mathbb{R} - \{0\} - \left[\frac{12}{31}, \frac{12}{29}\right]$

Comprehension 1: Q1) D Q2)C Q3)A

{Hint: 11) $|3x|$, $|3x - 1|$ and $|3x + 1|$ are in A.P.

$$\Rightarrow 2 \times |3x - 1| = |3x| + |3x + 1|$$

The critical points in this case are $\left\{-\frac{1}{3}, 0, \frac{1}{3}\right\}$



Case I : $x \in \left(-\infty, -\frac{1}{3}\right)$

In this case equation becomes

$$2 \times -(3x - 1) = -3x - (3x + 1)$$

$$\Rightarrow -6x + 2 = -6x - 1 \text{ which gives no solution}$$

Case II: $x \in \left[-\frac{1}{3}, 0\right)$

In this case, the equation becomes

$$2 \times (3x - 1) = -3x - (3x + 1)$$

$$\Rightarrow 6x - 2 = -6x - 1$$

$$\Rightarrow x = \frac{1}{12} \text{ (which is accepted as } \frac{1}{12} \in \left[-\frac{1}{3}, 0\right))$$

Case III : $x \in \left[0, \frac{1}{3}\right)$

In this case, the equation becomes

$$2 \times (3x - 1) = 3x - (3x + 1)$$

$$\Rightarrow 6x - 2 = -1$$

$$\Rightarrow x = \frac{1}{6} \text{ (which is rejected as } \frac{1}{6} \notin \left[0, \frac{1}{3}\right))$$

Case IV : $x \in \left[\frac{1}{3}, \infty\right)$

In this case, the equation becomes

$$2 \times (3x - 1) = 3x + (3x + 1)$$

i.e $6x - 2 = 6x + 1$. Hence no solution in this case also.

Combining all the cases, we get $x = \frac{1}{12}$

12) Substituting the value of x , the series becomes $\left\{\frac{1}{4}, \frac{3}{4}, \frac{5}{4}, \dots\right\}$

Hence, the series has first term $a = \frac{1}{4}$ and common difference $d = \frac{1}{2}$.

$$T_{10} = a + (10 - 1)d = \frac{1}{4} + 9 \times \frac{1}{2} = \frac{19}{2}$$

13) The sum of first 15 terms of the series is

$$S_{15} = \frac{n}{2} (2a + (n - 1)d) = \frac{15}{2} \left(2 \times \frac{1}{4} + 9 \times \frac{1}{2}\right) = \frac{15}{2} \times \frac{15}{2} = \frac{225}{4}$$

Comprehension2: Q1) A Q2)B Q3)C

$$\{\text{Hint: } f_1(x) = \frac{2x-1}{x+1} \text{ for } x \in \mathbb{R} - \{-1\}\}$$

$$f_2(x) = f_1(f_1(x)) = \frac{x-1}{x} \text{ for } x \in \mathbb{R} - \{-1, 0\}$$

$$f_3(x) = f_1(f_2(x)) = \frac{2-x}{1-2x} \text{ for } x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}\right\}$$

$$\text{On similar lines, we can find } f_4(x) = f_1(f_3(x)) = \frac{1}{1-x} \text{ for } x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}, 1\right\}$$

$$f_5(x) = f_1(f_4(x)) = \frac{1+x}{2-x} \text{ for } x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}, 1, 2\right\}$$

$$f_6(x) = f_1(f_5(x)) = x \text{ for } x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}, 1, 2\right\}$$

It may be noted that this function repeats itself after a gap of 6 .i.e $f_{6n+r}(x) = f_6(f_6 \dots n \text{ times}(f_r(x))) = f_r(x)$ for $x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}, 1, 2\right\}$

$$\text{Q14: } f_{34}(x) = f_4(x) \text{ for } x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}, 1, 2\right\}$$

$$\Rightarrow f_{34}(x) = \frac{1}{1-x} \text{ for } x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}, 1, 2\right\}$$

$\Rightarrow g(x) = |\ln(f_{34}(x))|$ has domain given by

$$\frac{1}{1-x} > 0 \text{ for } x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}, 1, 2\right\}$$

$$\text{i.e } x < 1 \text{ and } x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}, 1, 2\right\}$$

$$\text{i.e } x \in \left((-\infty, 1) - \left\{-1, 0, \frac{1}{2}\right\}\right)$$

$$\text{Q15: } |f_{71}(x)| > \left|\frac{1}{f_{73}(x)}\right|$$

$$\text{i.e we have to solve } |f_5(x)| > \left|\frac{1}{f_1(x)}\right| \text{ for } x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}, 1, 2\right\}$$

$$\text{i.e } \left|\frac{1+x}{2-x}\right| > \left|\frac{1}{\frac{2x-1}{x+1}}\right| \text{ for } x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}, 1, 2\right\}$$

$$\text{i.e } |2x-1| > |2-x| \text{ for } x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}, 1, 2\right\}$$

Squaring both sides we get

$$4x^2 - 4x + 1 > 4 + x^2 - 4x \text{ for } x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}, 1, 2\right\}$$

$$\text{i.e. } 3x^2 > 3 \text{ for } x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}, 1, 2\right\}$$

$$\Rightarrow x \in (\mathbb{R} - \{2\}) - [-1, 1]$$

$$\text{Q16: } |f_{56}(x)| > f_{56}(x)$$

$$\Rightarrow |f_2(x)| > f_2(x) \text{ for } x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}, 1, 2\right\}$$

$$\Rightarrow f_2(x) < 0 \text{ for } x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}, 1, 2\right\}$$

$$\Rightarrow \frac{x-1}{x} < 0 \text{ for } x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}, 1, 2\right\}$$

$$\Rightarrow x \in \left((0, 1) - \left\{\frac{1}{2}\right\}\right)$$

These values are a subset of $(0, 1)$.